Supplementary Information

Supplementary Text

1 Information regarding pre-registration and exploratory analyses

We first performed exploratory analyses using the popular music dataset. Based on the results of these analyses, we pre-registered the analyses we report for the unfamiliar music sample at https://osf.io/mw435.

We note one deviation from the pre-registration: The pre-registration states that we would select the latent emotion dimensions to retain after dimension reduction by using the elbow method on the singular values produced by *softImpute*[1]. The elbow method is a subjective visual heuristic. Instead, we opted for the more statistically rigorous approach reported in the manuscript, using Monte Carlo holdout cross-validation to determine the optimal parameters[2]. We implemented this approach using scripts from the R-miss-tastic ecosystem[3], purpose-built for parameter selection in *softImpute.*

2 Dimension reduction using SoftImpute

The data are sparse-by-design, with each of the 25 GEMS dimensions containing on average 11.11% (popular music corpus) and 11.7% (unfamiliar music corpus) complete responses. To address the sparse data structure via dimension reduction we used *softImpute*[1], a matrix completion method that assumes that listeners' ratings can be explained by a smaller set of latent factors, and provides a low-rank solution while estimating missing values[4]. As such, *softImpute* combines the steps of imputation and dimension reduction. *SoftImpute* was built to address the problem of sparse matrices, such as in the Netflix challenge: A recommendation algorithm must be able to use viewers' ratings for less than 2% of all content featured on the platform to predict ratings for the remaining unseen movies[4].

Our analyses focused on countries with at least 400 participants to balance sufficient data per country with maintaining a broad range of countries for further analysis (n = 19 countries for listeners who heard popular music samples, and n = 44 countries for listeners who heard unfamiliar music). Because *softImpute* does not natively handle nested data, we first separated systematic sources of variation (due to repeated measurements per country, song, and listener), from variance shared across the emotion ratings. To do so, we first fit linear mixed-effects models with random intercepts per country, listener, and song, for each of the 25 emotion dimensions. From each mixed-effects model, we extracted two sets of data: the predicted value on each of the 25 emotion dimensions based only on country, listener, and song information; and the residuals (where available), representing the remaining unexplained variation. This allowed us to model the systematic variance due to the grouping structure of the data, and then approximate the latent emotion structure in the residuals.

We then approximated the residual matrix as low-rank products of latent factors. To do so, we first selected the optimal regularisation parameter (λ) via Monte Carlo holdout cross-validation using scripts from the R-miss-tastic ecosystem[3]. Across 10 random splits, 10% of the observed entries were masked and *softImpute* was fitted with 20 candidate log-spaced λ values; we chose the λ value that minimised reconstruction error. We then imputed the residuals using this value, and re-constructed the full dataset by adding the imputed residuals back to the values predicted by the mixed-effects models. Finally, the completed dataset was projected onto the latent factors. This procedure reduced the 25 (incomplete) emotion rating dimensions to two (completed) latent factors.

We did not a-priori specify the number of latent components to retain. Using the optimal λ value, two latent components survived singular-value shrinkage, in both the popular music data and in the unfamiliar music data (see Figures S1a and S2a for loadings). These components can be interpreted similarly to principal components. We mean-centered and scaled data before subsequent analyses.

3 Replication analyses for primary models

3.1 Modeling with GEMS-9 factor solution

We also conducted the variance analysis with the raw, non-imputed data, with responses aggregated to produce the nine GEMS-9 factors[5] (see Methods), rather than reduced to a two-factor solution. Per Zentner and colleagues, the 25 emotion dimensions featured in the GEMS-25 can be reduced to the 9-factor solution by calculating the average of the 2-3 emotion labels that map onto each dimension. Following this, we computed GEMS-9 dimension scores for each listener on each song by averaging their available ratings on the relevant GEMS-25 dimensions. We then mean-centered and rescaled the resulting scores within each dimension to aid model fit.

To estimate the variance explained by country, listeners and song, we fit separate mixed models for each GEMS dimension. We then regressed ratings per dimension onto random intercepts for country, listener, and song. The variances attributable to country, listener, and song were significantly larger than zero (Random-effects ANOVAs; all ps $< .001$) for all nine dimensions, for both popular music and unfamiliar music.

These unimputed data are considerably noisier due to missingness and the inclusion of all 165 countries, including those with small sample sizes, in the models. Despite this noise, the pattern of results is consistent with the imputed data, with individual differences explaining the most variance in responses, followed by the song listeners heard, and finally the country the listener was from. For popular music, the individual listener explained between 8 and 28 times as much variance in responses as the country the listener was from (Figure S5a and Table S11). For unfamiliar music, the individual listener explained between 28 and 69 times as much variance in responses as the country the listener was from (Figure S5b and Table S12).

3.2 Restriction to large country-wise samples

The variance analyses reported in the manuscript use imputed and reduced data from countries that provided at least 400 listeners. To ensure that our results were not influenced by the choice of sample size for these models, we replicated the main imputation pipeline and analyses using data from countries with at least 800 listeners. This stricter limit resulted in n = 8 countries for listeners who heard popular music samples, and n = 29 countries for listeners who heard unfamiliar music. The imputation and dimension reduction procedure once again resulted in two dimensions tracking Valence and Arousal, visualised in Figure S8.

The results reported in the main manuscript replicated. For popular music, the ratio of within-country to between-country variability in emotion ratings was 11:1 valence and 7:1 for arousal. When analysing emotional responses to unfamiliar music, within-country variability again exceeded between-country variability (29:1 for valence and 25:1 for arousal).

3.3 Comparison to weather data

To put the variance decomposition results into context and rule out the results being an analytical artefact, we applied the same variance analyses to data where we *would* expect to see substantial country-level variation: weather data, which has also been used as a contrast to cross-country music datasets in prior work[6]. We used publicly available data (GSOY; Global Summary of the Year) provided by the NOAA National Centers for Environmental Information. This dataset also has high missingness (although unlike our data, weather station data are often missing non-randomly), and is similar in structure to our musical dataset: Weather stations, which are nested within countries, record multiple weather measures, such as temperature, rainfall and snowfall, across a range of years. This produces a similar structure to our dataset where listeners, nested within countries, provide multiple emotion ratings across a range of songs.

In contrast to our music dataset, the same analysis on weather data revealed that country-level variation explained 3 times as much variance in maximum yearly temperature as stations nested within countries

did, and 177 times as much as the year of recording did. Other measures, such as annual precipitation, demonstrated similar patterns (Figure S5c). The only exception was the number of days with snow; country-level variation explained 0.85 times as much variance in days with snow as stations nested within countries did. This reflects the fact that weather stations equipped to measure snow depth are more likely to be located in regions where snowfall is expected, resulting in selective sampling and less between-country variance.

3.4 Variance decomposition by language

Country is an imperfect proxy for culture. To ensure that this choice of measure was not driving results, we also replicated the variance decomposition analyses using another proxy for cultural similarity: language[7,8]. We ran these analyses both at the level of the native language reported by the listener, and also that language's broader language family according to Glottolog 5.3[9]. We fit separate mixed models for each of the GEMS-9 dimensions and regressed ratings per dimension onto random intercepts for listener, song and language (or language family). The results were consistent with the country-based analyses: individual differences explained the most variance in responses, followed by the song listeners heard, and finally the listener's native language or language family that the language belonged to.

For popular music, the median ratio of within-language variance to between-language variance was 26:1 (range: 8 - 40; Table S13). The median ratio of variance within a language family to between families was 19:1 (range: 7 - 34; Table S14). This pattern persisted even when excluding the largest language family represented in our dataset, Indo-European, which was dominated by English-speakers (median ratio: 20:1, range: 8 - 52).

For unfamiliar music, the ratio of within-group variance to between-group variance were higher. The median ratio of within-language variance to between-language variance was 51:1 (range: 30 - 88; Table S15), while the median ratio of variance within a language family to between families was 44:1 (range: 24 - 98; Table S16).

4 Supplementary analyses for correlation analysis

4.1 Noise ceiling correction

In the main manuscript, we report correlations between the mean song-wise ratings of pairs of countries. These correlations are likely underestimates of the true correlation due to the sparseness of the data and resulting sampling error. For example, even if two countries' song-wise ratings on *peacefulness* are perfectly correlated in reality, the noise present within each country's data will limit the maximum possible observable correlation using these data. A noise ceiling reflects an estimate of the maximum observable correlation given the noise in the data. As suggested by Cowen and colleagues[10] and following the same process described in[11], we corrected for this upper limit by calculating the noise ceiling of each country on each GEMS-9 dimension, which represents an upper limit for correlations using the data from the country.

We calculated the noise ceilings using the following formula:

$$\sqrt{1 - \frac{mean(\text{squared SE per song on a given rating dimension})}{var(\text{mean score per song on a given rating dimension})}}$$

We then adjusted each pairwise correlation by dividing the observed correlation coefficient by the average of the noise ceilings in that pair of countries (the distribution of adjusted correlation coefficients is visualised in Figure S4). This provides a measure of how the observed correlation compares to the average "expected" maximum correlation, if the data in both countries were perfectly reliable. As it is technically possible for the adjustment to result in a value above r = 1, we capped any values that exceeded this. In the Western Popular Music sample, the average adjusted correlation coefficients ranged from r = 0.34 (for *transcendence*)

to r = 0.73 (for *peacefulness*), with an average r of 0.57 across all nine dimensions. In the unfamiliar music sample, the average adjusted correlation coefficients ranged from r = 0.57 (for *transcendence*) to r = 0.89 (for *joyful activation*), with an average r of 0.75.

5 Information regarding linked datasets

5.1 The original behavioural contexts for unfamiliar music

The stimuli that comprised the unfamiliar music corpus were the 118 songs featured in the Natural History of Song (NHS) Discography[6]. The NHS Discography features 14-second song segments of archival recordings of songs from 86 mostly smaller-scale societies. These societies span 30 world regions[12,13], over 75 languages, and a range of subsistence methods, resulting in a highly diverse set of world music.

The songs featured in the Discography were originally used in four behavioural contexts: soothing a baby, dancing, expressing love, and healing illness. Researchers determined the original behavioural contexts of the songs using accompanying liner notes and supporting ethnographic texts, without first listening to the audio. All excerpts and their metadata can be downloaded from https://doi.org/10.5281/zenodo.7265514.

To test whether listener's emotional responses to music reflected the songs' original functions, we calculated the mean arousal and valence rating per song across all participants, and regressed these ratings onto song type. We found that how the songs were originally used in the cultures where they were recorded did indeed predict the emotions listeners reported in this study (Figure S6). The largest effects were on the arousal dimension. Both dance songs and lullabies differed from the mean arousal rating, in opposite directions: Dance songs were higher, and lullabies lower than average in terms of perceived arousal (dance: β = 0.34, $p < .0001$; lullabies: β = -0.33, $p < .0001$). Lullabies were rated lower on arousal than both dance songs (β = -0.66, $p < .0001$) and healing songs (β = -0.42, $p < .001$), while dance songs were also higher arousal than love songs (β = 0.43, $p < .001$). Only dance songs differed from the mean valence score, positively (β = 0.18, p = 0.02); dance songs and lullabies were distinguishable from each other based on this dimension (β = -0.29, p = 0.05).

5.2 Perceived behavioural contexts for unfamiliar music

In the manuscript, we analyse the relationship between induced emotions, as reported by participants in the present study, and the perceived behavioural context of the songs in the unfamiliar music sample. The latter ratings come from the publicly available data analysed in Yurdum and colleagues'[11] work, accessible at https://github.com/themusiclab/universal-music. This dataset contains ratings from two samples of listeners: 5,516 listeners from 45 industrialised countries (tested online in the non-English native language of their country), and 116 listeners from three smaller-scale societies: the Nyangatom in Ethiopia, the Mentawai in Indonesia, and the Tannese Ni-Vanuatu in Vanuatu.

In both samples, participants heard a song excerpt from the NHS Discography and then rated what they thought the song was originally used for. Participants from industrialised countries were prompted with the text "Think of the people making this music. I think that they…" and asked to respond on a scale from 1 ("Definitely do not use the music… [context]") to 4 ("Definitely use the music… [context]"), where [context] referred to each of the four behavioural contexts represented in the corpus: "for dancing," "to soothe a baby," "to heal illness," and "to express love for another person." In the smaller-scale societies, the experimenter read instructions out loud in the participant's native language. The prompt was modified to "Do you think they use the music for [context]?" with the response options "no," "a little," and "yes". Ratings from all participants were centered and scaled within their own cohorts. Higher scores indicate that a song excerpt was judged as being more appropriate for the tested context. For more information regarding the study procedure, translations, and results, see[11].

We asked whether the emotion ratings gathered in the present manuscript were related to the perceived behavioural contexts of the songs, collected from participants who spoke non-English native languages,

in both non-globalised communities and globalised countries[11]. They were; in almost all cases listeners' emotional responses were strongly related to the perceived functions of the songs as rated by internet-connected participants in their native languages (Figure S7a). High-valence songs were perceived as more appropriate for dancing (β =1.19, SE =0.18, $p < .0001$) and expressing love for another person (β =1.48, SE =0.16, $p < .0001$), and less appropriate for healing illness (β =-1.26, SE =0.18, $p < .0001$). Arousal predicted how appropriate listeners thought a song was for dancing (β =1.42, SE =0.14, $p < .0001$), and how *in*appropriate it would be for soothing infants (β =-1.62, SE =0.12, $p < .0001$) or expressing love for another person (β =-1.06, SE =0.17, $p < .0001$). The emotional ratings were also strongly related to the perceived functions of the songs as rated by the listeners from three relatively isolated cultures (Figure S7b): High-valence songs were rated as more appropriate for dancing (β =0.99, SE =0.19, $p < .0001$) and expressing love for another person (β =0.94, SE =0.20, $p < .0001$), while high-arousal songs were rated as appropriate for dancing (β =1.23, SE =0.16, $p < .0001$) and inappropriate for soothing infants (β =-1.16, SE =0.16, $p < .0001$).

These results suggest correspondence between the emotional judgements made by globalised participants in this English-language experiment, and perceptual judgements made about the same songs in prior work, despite the fact that the listeners in prior work were not primarily English speakers, and in some cases, not from globalised communities.

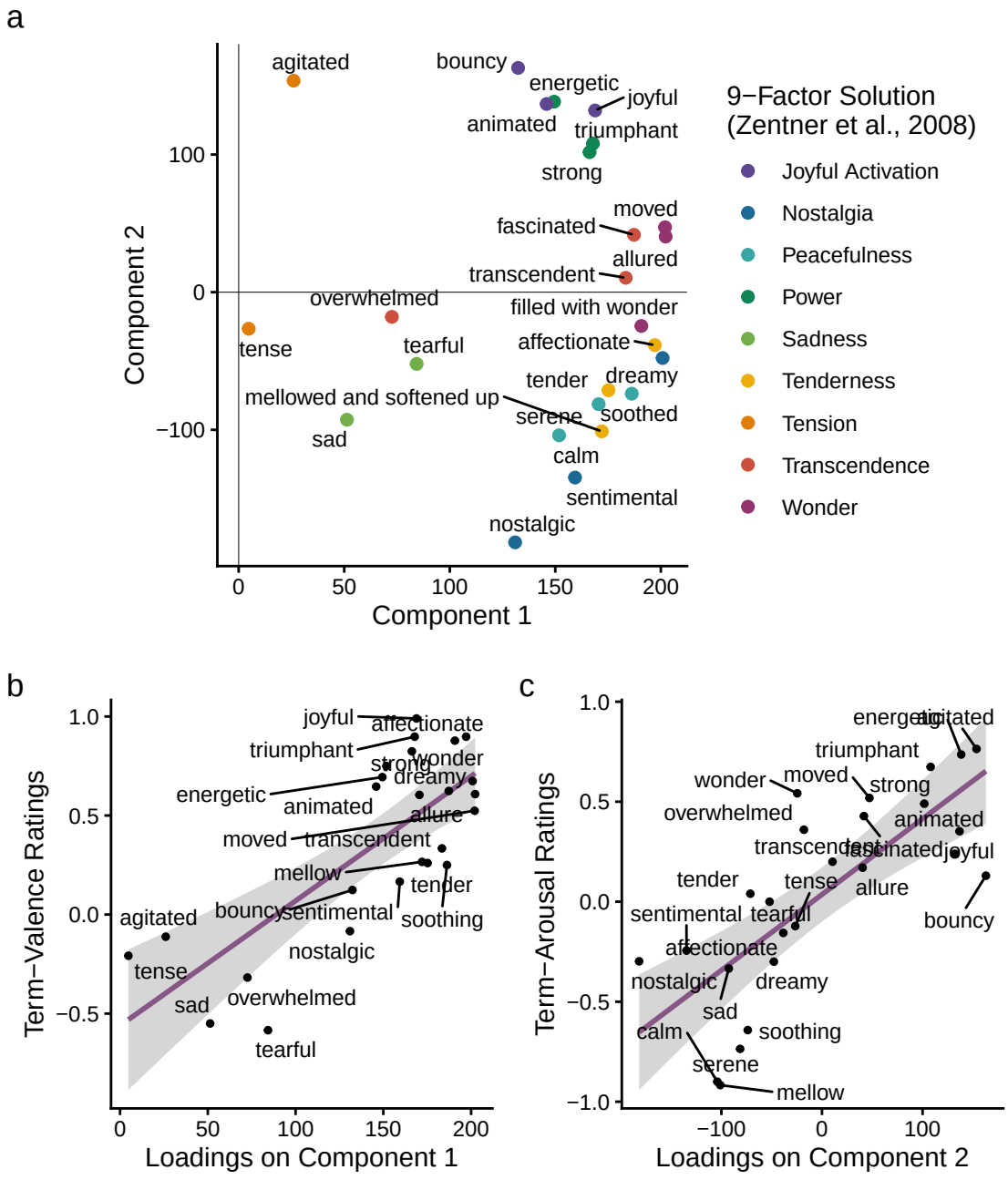


Figure S1. Results of dimension reduction procedure on emotion ratings of popular music. Panel (a) plots the 25 GEMS emotion ratings in reduced 2D space. Points indicate the original variables that contributed to the imputed low-rank structure, colour-coded in line with Zentner and colleagues' GEMS-9 solution. To interpret the two components produced by dimension reduction, we correlated the loadings of each emotion term on the component with the valence and arousal ratings of the same terms in the NRC VAD lexicon[14]. The emotion terms that loaded most strongly on Component 1 were high in valence (a), whereas Component 2 was characterised by high-arousal terms (b).

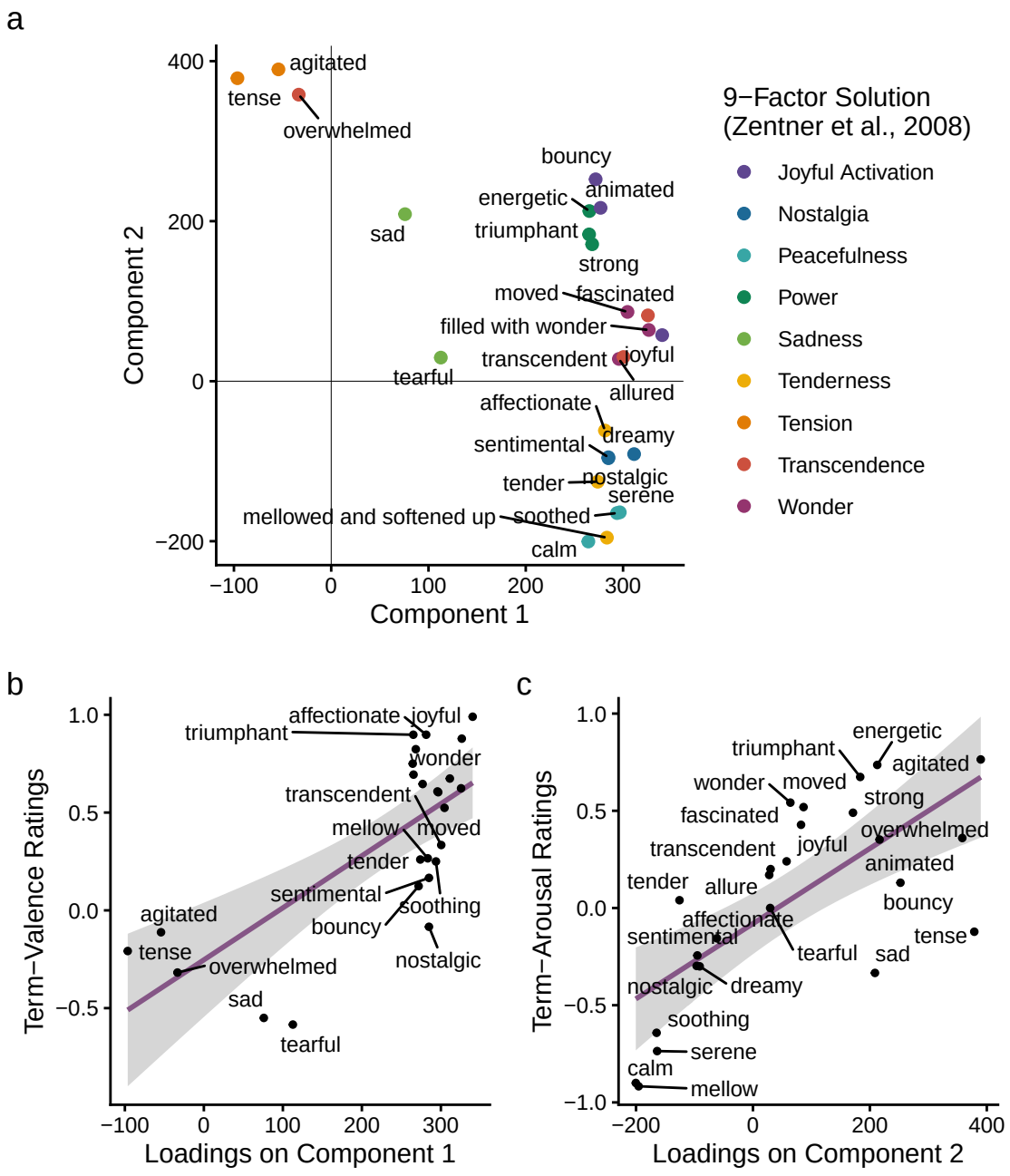


Figure S2. Results of dimension reduction procedure on emotion ratings of unfamiliar music. Panel (a) plots the 25 GEMS emotion ratings in reduced 2D space. Points indicate the original variables that contributed to the imputed low-rank structure, colour-coded in line with Zentner and colleagues' GEMS-9 solution. To interpret the two components produced by dimension reduction, we correlated the loadings of each emotion term on the component, with the valence and arousal ratings of the same terms in the NRC VAD lexicon[14]. The emotion terms that loaded most strongly on Component 1 were high-valence (a), whereas Component 2 was characterised by high-arousal terms (b).

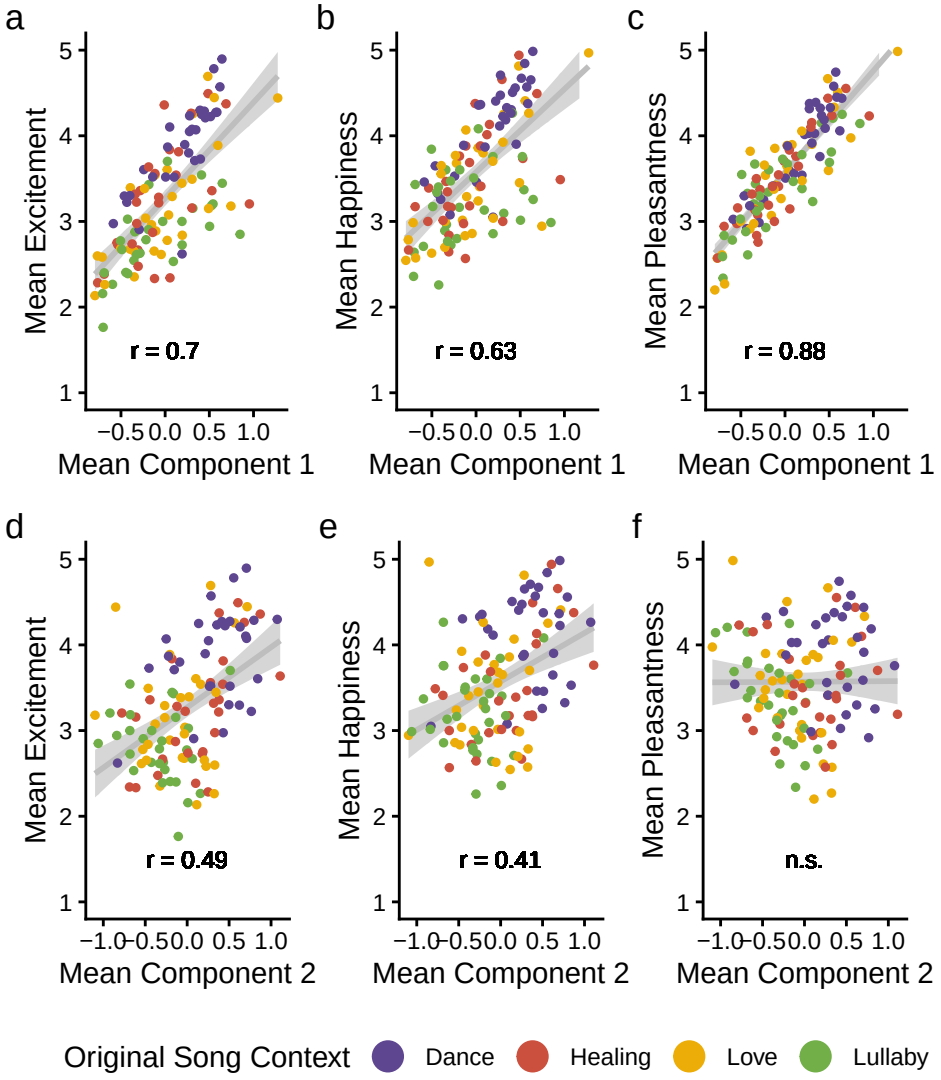


Figure S3. Song-wise mean ratings on the reduced components correlated with ratings of pleasantness and excitement from prior work. Prior work by Mehr and colleagues[15] contained ratings of 'happiness', 'pleasantness', and 'excitement' for the 118 unfamiliar music samples we used in the present study. These ratings were intended as proxies for valence and arousal. We used these ratings to validate our interpretation of Component 1 as tracking valence, and Component 2 as tracking arousal. The song-wise mean scores on Component 1 were most strongly correlated with song-wise mean ratings of pleasantness (c), while Component 2 correlated most strongly with ratings of excitement (d). All correlations were significant at $p < .0001$, except between Component 2 and pleasantness (f). The colour of the dot indicates the original behavioural context the song was performed in.

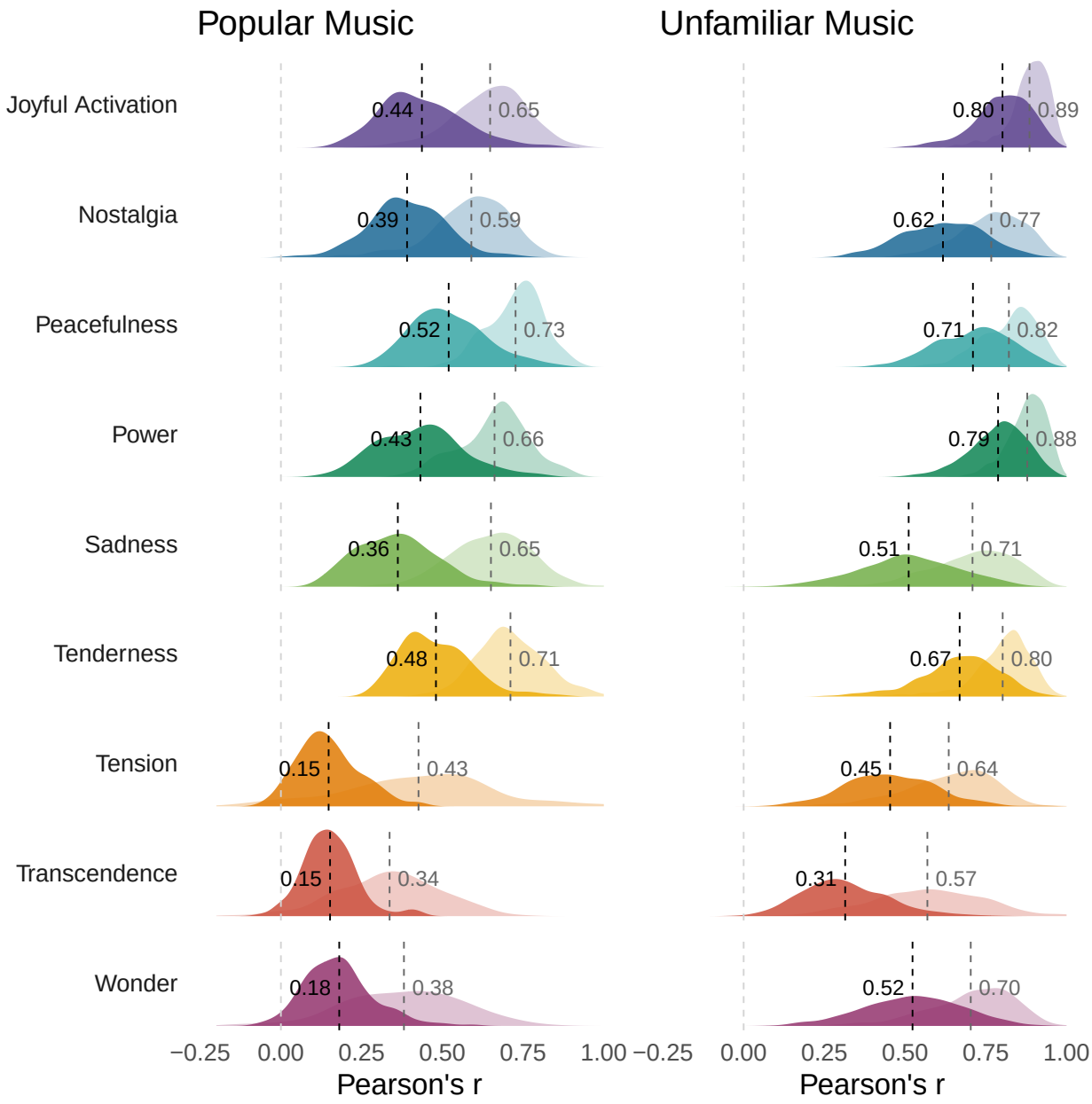


Figure S4. Emotional responses to music are correlated across countries. The curves represent the distribution of Pearson's r correlation coefficients observed between countries with at least 400 participants, on each of the nine GEMS dimensions, for popular and unfamiliar music. The lighter curves represent the distribution of correlation coefficients after applying a noise ceiling correction to each one, described in SI Text 4.1, to adjust for the sampling error present in the data.

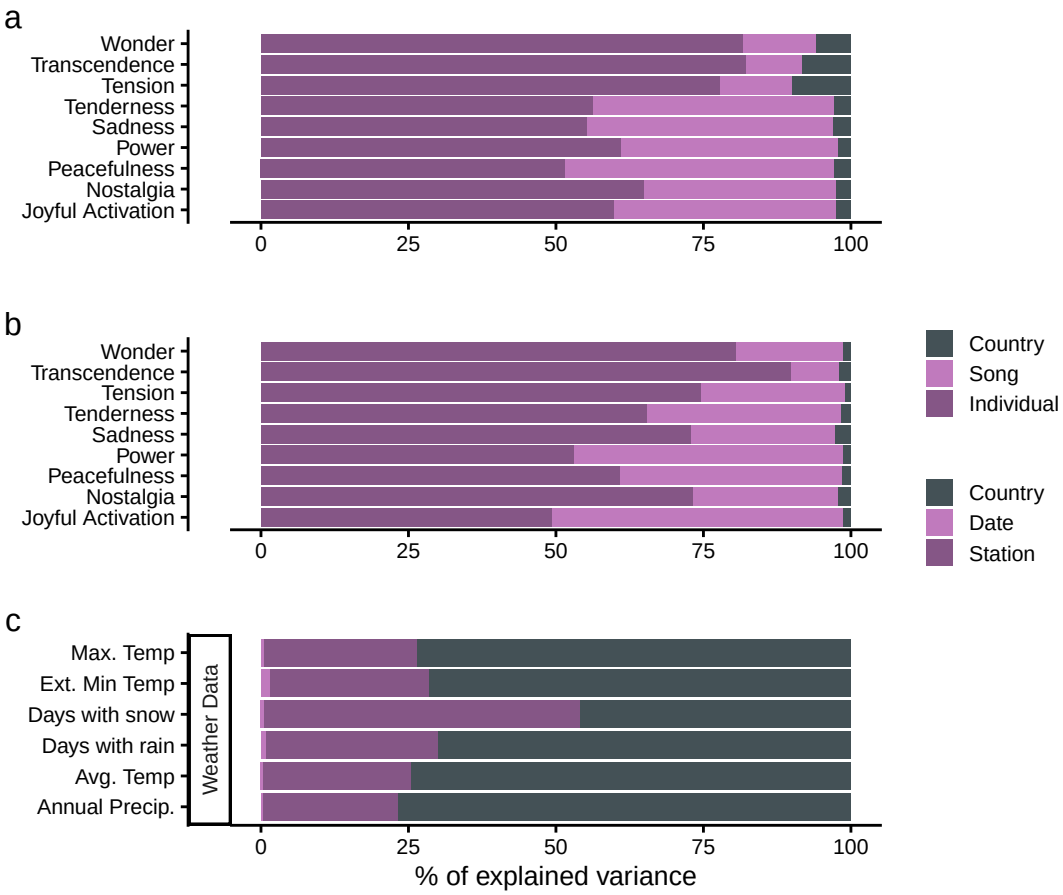


Figure S5. Emotional responses to music vary considerably more among individuals and across songs than they do across countries. Proportion of explained variance attributable to the country the listener lives in, the song they heard, and the listener's own unique response pattern for a) popular music, and b) unfamiliar music. Panel (c) contrasts these results with various measures of weather characteristics from the Global Summary of the Year. Almost all weather features vary substantially more from country to country than within a single country. (Avg. Temp = Average annual temperature; Annual Precip. = Total annual precipitation; Days with rain = Number of days with >= 0.01 inch/0.254 millimetre rain in the year; Days with snow = Number of days with snow depth >= 1 inch/25 millimetres.)

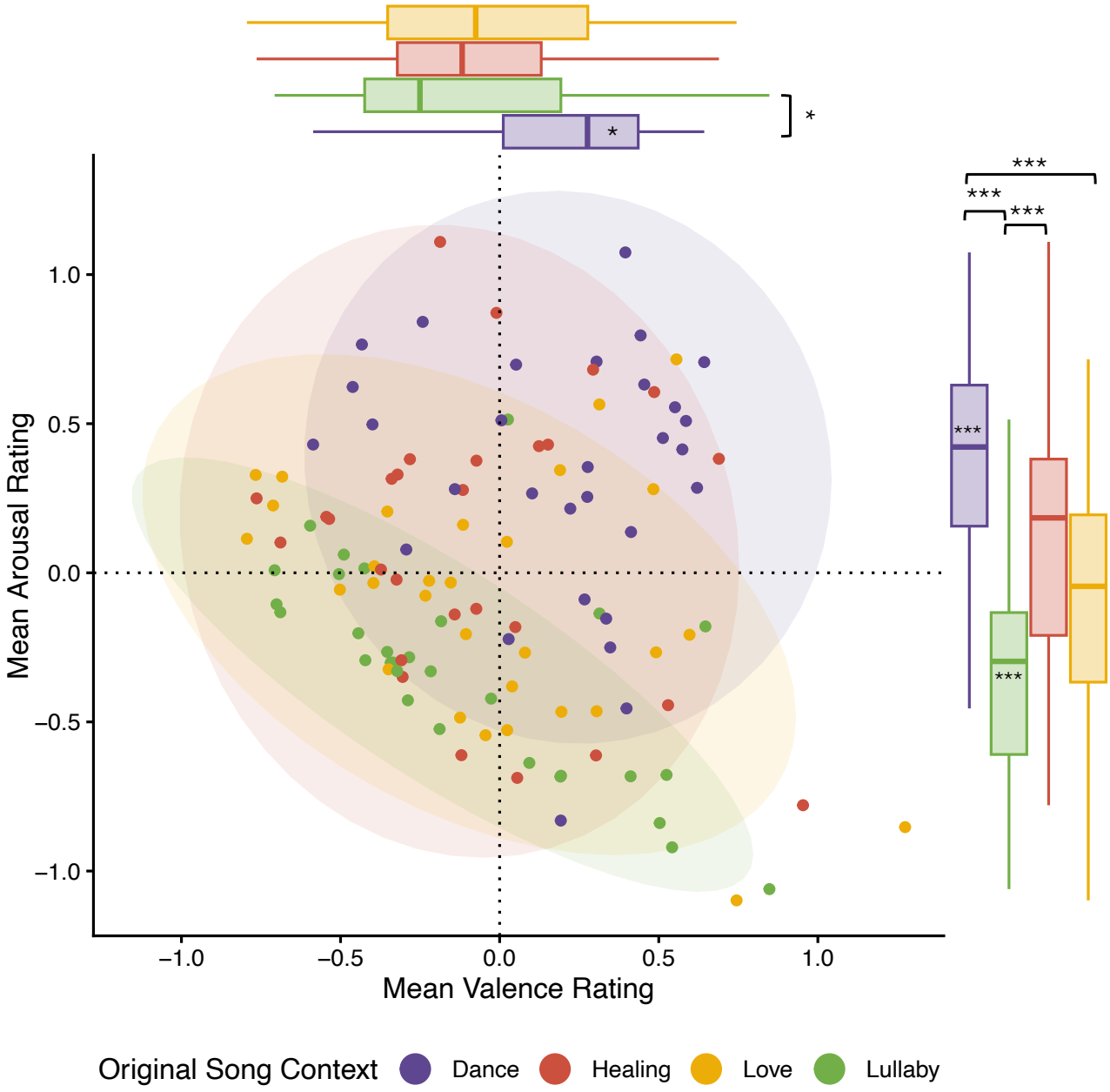


Figure S6. The original behavioural context of a song predicts the emotions it induces. Each dot represents the average Valence and Arousal score of a song from the unfamiliar music corpus, as rated by participants. The colour of the dot indicates the original behavioural context in which each song was performed in the (non-English speaking) society it was recorded in: soothing infants, healing illness, dance, or expressing love for another person. Significance stars within the boxplots denote a difference from the mean rating across all songs (0); significance stars next to brackets denote pairwise comparisons between types of songs. *** $< .001$, ** $< .01$, * $< .05$

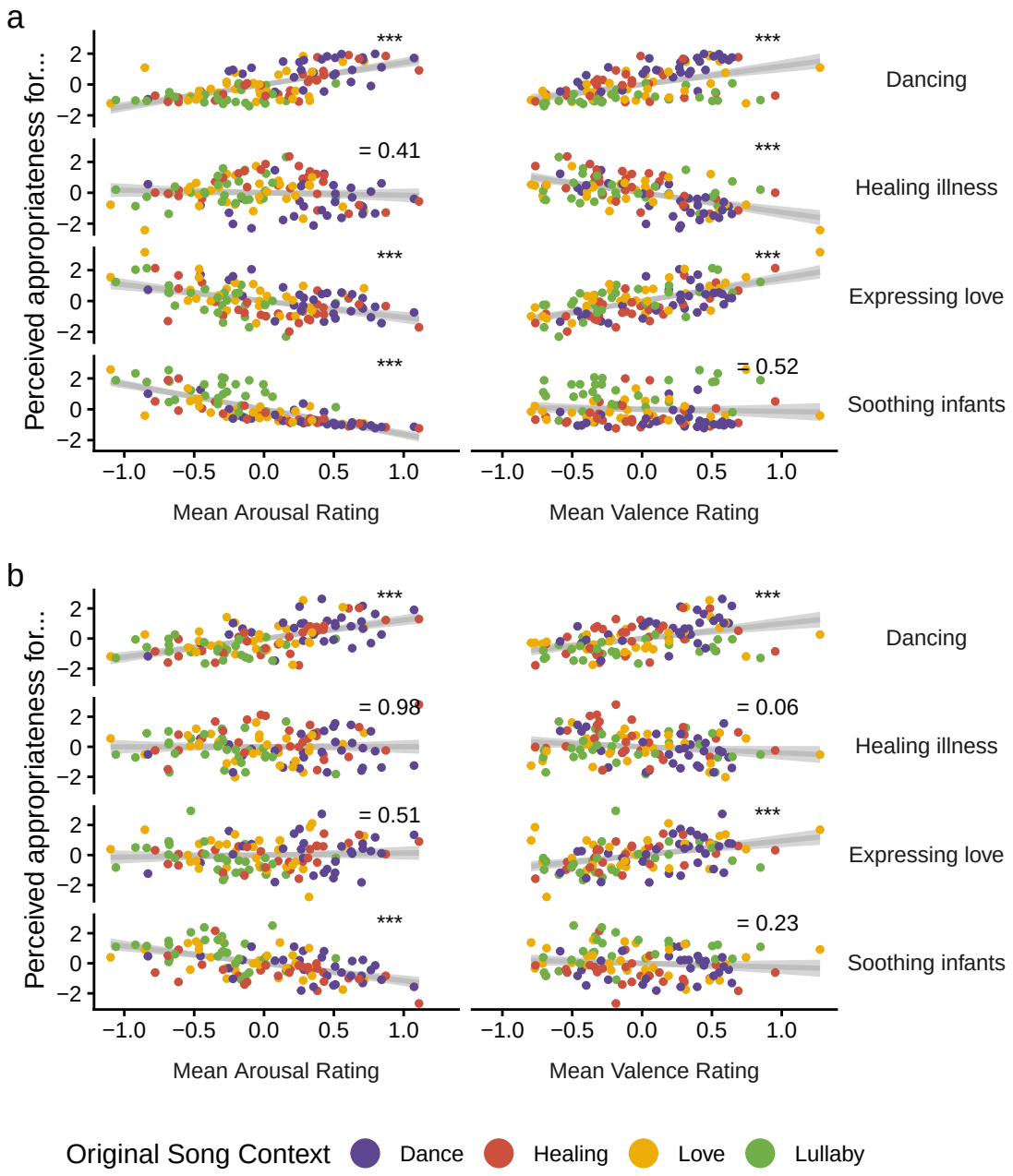


Figure S7. The emotions induced by songs predicted their perceived behavioural contexts. The x axes represent the average valence or arousal score of a song from the unfamiliar music corpus, as rated by participants in the present study. The y axes indicate the perceived appropriateness of each song in one of four behavioural contexts: dancing, expressing love for another person, healing illness, or soothing infants. These ratings are taken from Yurdum and colleagues' work[11] and reflect the perceptions of (a) listeners from 45 industrialised countries, tested in their non-English native languages, and (b) listeners from three smaller-scale societies. The colour of the dot indicates the original behavioural context the song was performed in. All tests marked with *** are significant at $p < .001$. Non-significant p values are presented in text.

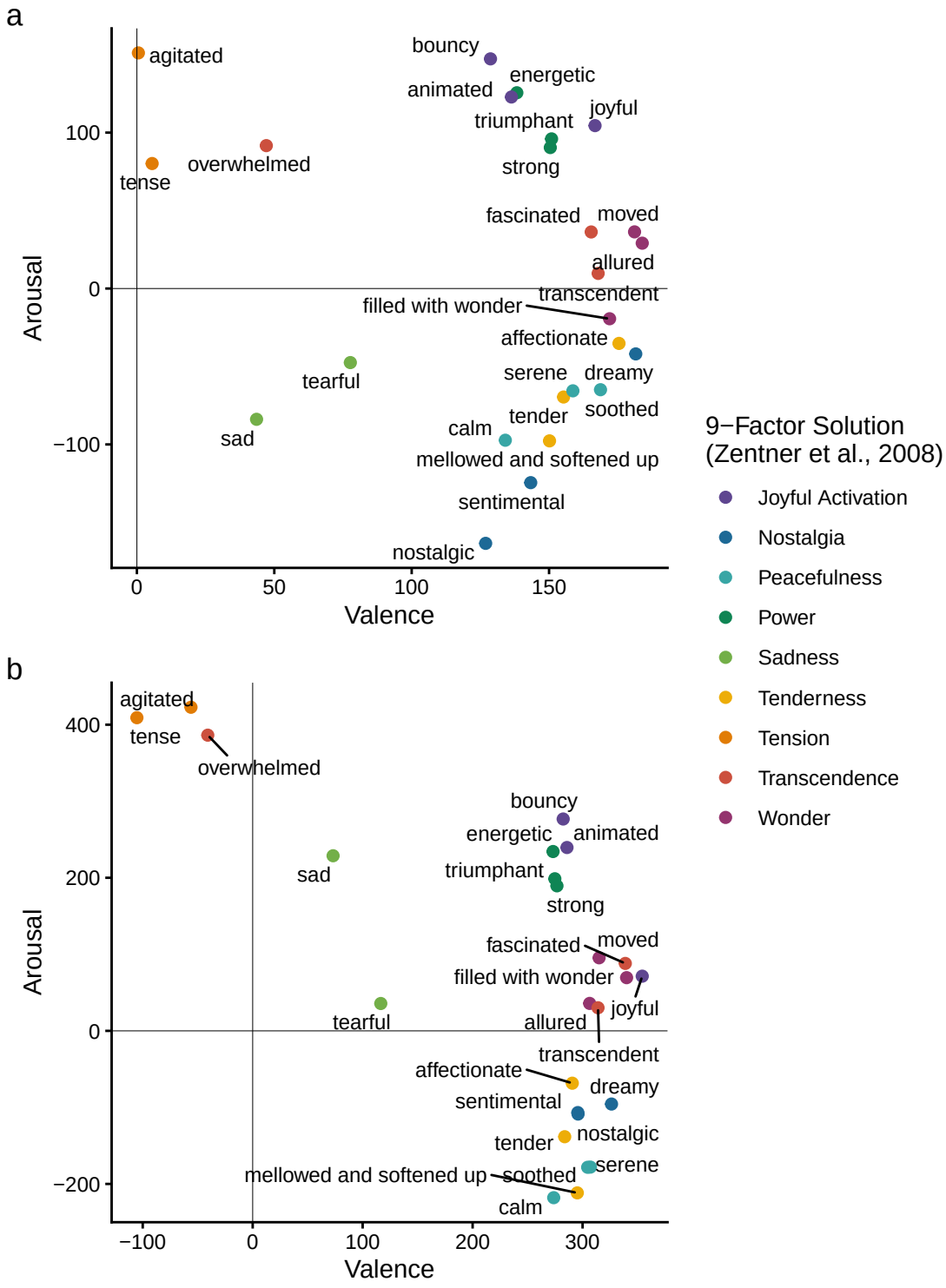


Figure S8. Singular vector loadings for countries with over 800 participants. Dots indicate the original variables that contributed to the imputed low-rank structure, colour-coded in line with the GEMS-9 factors. Panels (a) and (b) show the loadings produced after filtering the data to countries with over 800 participants, for the popular music sample and the unfamiliar music sample, respectively.

Sample	GEMS-9 Dimension	Mean r	Median r (IQR)	Adjusted mean r
Popular Music	joyful activation	0.44	0.42 (0.17)	NA
	nostalgia	0.39	0.38 (0.15)	0.59
	peacefulness	0.52	0.51 (0.15)	0.73
	power	0.43	0.44 (0.18)	0.66
	sadness	0.36	0.36 (0.18)	0.65
	tenderness	0.48	0.46 (0.15)	0.71
	tension	0.15	0.13 (0.12)	0.43
	transcendence	0.15	0.15 (0.10)	0.34
	wonder	0.18	0.17 (0.13)	0.38
Unfamiliar Music	joyful activation	0.80	0.81 (0.11)	NA
	nostalgia	0.62	0.62 (0.18)	0.77
	peacefulness	0.71	0.72 (0.17)	0.82
	power	0.79	0.80 (0.12)	0.88
	sadness	0.51	0.51 (0.21)	0.71
	tenderness	0.67	0.68 (0.15)	0.80
	tension	0.45	0.45 (0.20)	0.64
	transcendence	0.31	0.30 (0.18)	0.57
	wonder	0.52	0.53 (0.21)	0.70

Table S1.

Pearson's r across the song-wise emotion ratings of all pairs of countries with at least 400 listeners. The mean r indicates a lower bound on the average pairwise correlation; the table also presents the median r and inter-quartile range (IQR) of the distribution of r values. The adjusted mean r indicates an estimate of the mean correlation after correcting each correlation in the distribution using a noise ceiling adjustment.

Sample	GEMS-9 Dimension	Mean r	Median r (IQR)
Popular Music	joyful activation	0.49	0.47 (0.17)
	nostalgia	0.44	0.44 (0.20)
	peacefulness	0.57	0.56 (0.18)
	power	0.49	0.49 (0.22)
	sadness	0.40	0.38 (0.20)
	tenderness	0.52	0.50 (0.15)
	tension	0.18	0.15 (0.15)
	transcendence	0.19	0.17 (0.12)
	wonder	0.20	0.19 (0.16)
Unfamiliar Music	joyful activation	0.79	0.80 (0.14)
	nostalgia	0.60	0.62 (0.22)
	peacefulness	0.70	0.72 (0.20)
	power	0.79	0.80 (0.11)
	sadness	0.53	0.52 (0.23)
	tenderness	0.68	0.69 (0.19)
	tension	0.44	0.44 (0.20)
	transcendence	0.31	0.29 (0.19)
	wonder	0.51	0.52 (0.26)

Table S2.

Pearson's r across the song-wise emotion ratings of all pairs of languages with at least 400 listeners. The mean r indicates a lower bound on the average pairwise correlation; the table also presents the median r and inter-quartile range (IQR) of the distribution of r values.

Factor	Group	Prop. of Variance
Valence	Listener	0.30
	Song	0.11
	Country	0.02
Arousal	Listener	0.05
	Song	0.42
	Country	0.01

Table S3.

Proportion of variance explained by the country the listener lives in, the song they heard, and the listener's own unique response pattern in response to popular music. These models use imputed and reduced data, which resulted in two components.

Factor	Group	Prop. of Variance
Valence	Listener	0.31
	Song	0.20
	Country	0.01
Arousal	Listener	0.11
	Song	0.23
	Country	0.00

Table S4.

Proportion of variance explained by the country the listener lives in, the song they heard, and the listener's own unique response pattern in response to unfamiliar music. These models use imputed and reduced data, which resulted in two components.

Demographic variable	Response	n	%
Race of participant	White	24,508	54.40
	Black or African American	1,325	2.94
	Asian	6,556	14.55
	Native Hawaiian or other Pacific Islander	191	0.42
	American Indian/Alaska Native	402	0.89
	More than one race	3,317	7.36
	Prefer not to say	2,401	5.33
	(Missing)	6,353	14.10
Is participant Latino?	Yes	4,473	9.93
	No	32,395	71.90
	Prefer not to say	1,814	4.03
	(Missing)	6,371	14.14
Participant's household income	Under $10,000	2,033	4.51
	$10,000 to $19,999	1,634	3.63
	$20,000 to $29,999	2,046	4.54
	$30,000 to $39,999	2,206	4.90
	$40,000 to $49,999	2,058	4.57
	$50,000 to $74,999	4,192	9.30
	$75,000 to $99,999	3,396	7.54
	$100,000 to $150,000	4,792	10.64
	Over $150,000	3,900	8.66
	I'd prefer not to say	12,190	27.06
	(Missing)	6,606	14.66

Table S5.

Demographic information for US-based participants, across both popular music and unfamiliar music corpora (n = 45,053).

Country	n
USA	14,357
Canada	1,923
UK	1,821
Germany	1,610
Turkey	1,312
France	1,153
Australia	1,031
Brazil	808
Poland	758
India	681
Russia	633
Spain	633
Argentina	627
Italy	582
Netherlands	545
Mexico	538
Sweden	452
Indonesia	440
Philippines	434

Table S6.

Countries that were reported as the native country of at least 400 participants who rated popular music.

Language	n
English	18,886
Spanish	3,063
German	1,700
French	1,544
Turkish	1,319
Portuguese	1,149
Russian	897
Polish	794
Chinese/Mandarin	621
Dutch	617
Italian	595
Indonesian	436
Swedish	418

Table S7.

Languages that were reported as the native language of at least 400 participants who rated popular music.

Country	n
USA	44,844
Canada	6,295
UK	5,764
Turkey	5,091
Germany	5,033
Australia	3,281
Spain	2,957
Poland	2,943
France	2,940
Argentina	2,601
Mexico	2,047
Brazil	2,006
Sweden	1,687
Netherlands	1,677
Italy	1,570
Russia	1,556
Singapore	1,535
Indonesia	1,463
India	1,445
Philippines	1,288
Chile	1,210
China	1,137
Finland	1,110
Norway	1,046
Colombia	890
Hong Kong	877
Portugal	872
Malaysia	833
Belgium	825
New Zealand	785
Denmark	782
Switzerland	774
Peru	745
Austria	703
Taiwan	695
Czech Republic	679
Greece	656

Romania	619
Hungary	588
Thailand	558
Ireland	543
Vietnam	503
Israel	423
Ukraine	402

Table S8.

Languages that were reported as the native language of at least 400 participants who rated unfamiliar music.

Language	n
English	60,312
Spanish	11,813
German	5,281
Turkish	5,160
French	3,990
Polish	3,155
Portuguese	2,984
Russian	2,240
Chinese/Mandarin	2,155
Dutch	1,857
Italian	1,639
Swedish	1,589
Indonesian	1,493
Chinese/Cantonese/Yue	1,231
Finnish	1,008
Tagalog	1,006
Norwegian	963
Greek	737
Romanian	719
Arabic	714
Danish	698
Hungarian	683
Vietnamese	667
Czech	642
Thai	539
Hindi	502
Korean	421
Chinese/Taiwanese/Min Nan/Hokkien	406

Table S9.

Languages that were reported as the native language of at least 400 participants who rated unfamiliar music.

Year	Artist	Song Name
2019	Ed Sheeran and Justin Bieber	I Don't Care
2019	Lil Nas X	Old Town Road
2019	Lizzo	Truth Hurts
2019	Drake	Money In The Grave
2018	Drake	God's Plan
2018	Ed Sheeran	Perfect
2018	Khalid and Normani	Love Lies
2018	Ariana Grande	No Tears Left to Cry
2017	Ed Sheeran	Shape of You
2017	Luis Fonsi and Daddy Yankee ft. Justin Bieber	Despacito (Remix)
2017	Rae Sremmurd ft. Gucci Mane	Black Beatles
2017	The Weeknd ft. Daft Punk	Starboy
2016	Justin Bieber	Love Yourself
2016	Justin Bieber	Sorry
2016	G-Eazy and Bebe Rexha	Me, Myself & I
2016	Twenty One Pilots	Ride
2015	Mark Ronson ft. Bruno Mars	Uptown Funk
2015	Ed Sheeran	Thinking Out Loud
2015	Skrillex and Diplo ft. Justin Bieber	Where Are U Now
2015	Rachel Platten	Fight Song
2014	Pharrell Williams	Happy
2014	Katy Perry ft. Juicy J	Dark Horse
2014	Passenger	Let Her Go
2014	Lorde	Royals
2013	Macklemore and Ryan Lewis ft. Wanz	Thrift Shop
2013	Robin Thicke ft. T.I. and Pharrell Williams	Blurred Lines
2013	Avicii	Wake Me Up
2013	Justin Timberlake ft. Jay-Z	Suit & Tie
2012	Gotye ft. Kimbra	Somebody That I Used to Know
2012	Carly Rae Jepsen	Call Me Maybe
2012	Train	Drive By
2012	Drake ft. Lil Wayne	The Motto
2011	Adele	Rolling in the Deep
2011	LMFAO ft. Lauren Bennett and GoonRock	Party Rock Anthem
2011	Pink	F**kin' Perfect
2011	Rihanna ft. Drake	What's My Name?
2010	Kesha	Tik Tok
2010	Lady Antebellum	Need You Now
2010	Mike Posner	Cooler Than Me
2010	The Black Eyed Peas	Imma Be
2009	The Black Eyed Peas	Boom Boom Pow
2009	Lady Gaga	Poker Face
2009	Soulja Boy Tell 'Em ft. Sammie	Kiss Me thru the Phone
2009	Jay Sean ft. Lil Wayne	Down
2008	Flo Rida ft. T-Pain	Low
2008	Leona Lewis	Bleeding Love
2008	Chris Brown ft. T-Pain	Kiss Kiss
2008	Ne-Yo	Closer
2007	Beyoncé	Irreplaceable
2007	Rihanna ft. Jay-Z	Umbrella
2007	Fergie ft. will.i.am	Fergalicious
2007	Soulja Boy Tell 'Em	Crank That (Soulja Boy)
2006	Daniel Powter	Bad Day
2006	Sean Paul	Temperature
2006	Rihanna	SOS
2006	Panic! at the Disco	I Write Sins Not Tragedies
2005	Mariah Carey	We Belong Together
2005	Gwen Stefani	Hollaback Girl
2005	The Game ft. 50 Cent	How We Do
2005	Weezer	Beverly Hills
2004	Usher ft. Lil Jon and Ludacris	Yeah!
2004	Usher	Burn
2004	Evanescence	My Immortal

(continued)

Year	Artist	Song Name
2004	Lil' Flip ft. Lea	Sunshine
2003	50 Cent	In Da Club
2003	R. Kelly	Ignition (Remix)
2003	Uncle Kracker ft. Dobie Gray	Drift Away
2003	Lil' Kim ft. 50 Cent	Magic Stick
2002	Nickelback	How You Remind Me
2002	Ashanti	Foolish
2002	Eve ft. Alicia Keys	Gangsta Lovin'
2002	Creed	My Sacrifice
2001	Lifehouse	Hanging by a Moment
2001	Alicia Keys	Fallin'
2001	Uncle Kracker	Follow Me
2001	112	Peaches & Cream
2000	Faith Hill	Breathe
2000	Santana ft. Rob Thomas	Smooth
2000	Christina Aguilera	What a Girl Wants
2000	Brian McKnight	Back at One
1999	Cher	Believe
1999	TLC	No Scrubs
1999	Santana ft. Rob Thomas	Smooth
1999	TLC	Unpretty
1998	Next	Too Close
1998	Brandy and Monica	The Boy Is Mine
1998	Puff Daddy ft. The Notorious B.I.G. and Mase	Been Around the World
1998	Sarah McLachlan	Adia
1997	Elton John	Something About the Way You Look Tonight / Candle in the Wind 1997
1997	Jewel	Foolish Games / You Were Meant for Me
1997	Az Yet ft. Peter Cetera	Hard to Say I'm Sorry
1997	The Notorious B.I.G. ft. Puff Daddy and Mase	Mo Money Mo Problems
1996	Los del Rio	Macarena (Bayside Boys Mix)
1996	Mariah Carey and Boyz II Men	One Sweet Day
1996	Eric Clapton	Change the World
1996	LL Cool J	Hey Lover
1995	Coolio ft. L.V.	Gangsta's Paradise
1995	TLC	Waterfalls
1995	Nicki French	Total Eclipse of the Heart
1995	Des'ree	You Gotta Be
1994	Ace of Base	The Sign
1994	All-4-One	I Swear
1994	Prince	The Most Beautiful Girl in the World
1994	Coolio	Fantastic Voyage
1993	Whitney Houston	I Will Always Love You
1993	Tag Team	Whoomp! (There It Is)
1993	Janet Jackson	If
1993	SWV	I'm So Into You
1992	Boyz II Men	End of the Road
1992	Sir Mix-a-Lot	Baby Got Back
1992	Michael Jackson	Remember the Time
1992	CeCe Peniston	Finally
1991	Bryan Adams	(Everything I Do) I Do It for You
1991	Color Me Badd	I Wanna Sex You Up
1991	Janet Jackson	Love Will Never Do (Without You)
1991	Marky Mark and the Funky Bunch	Good Vibrations
1990	Wilson Phillips	Hold On
1990	Roxette	It Must Have Been Love
1990	Wilson Phillips	Release Me
1990	Linda Ronstadt and Aaron Neville	Don't Know Much
1989	Chicago	Look Away
1989	Bobby Brown	My Prerogative
1989	Bobby Brown	On Our Own
1989	Phil Collins	Two Hearts

(continued)

Year	Artist	Song Name
1988	George Michael	Faith
1988	INXS	Need You Tonight
1988	Def Leppard	Pour Some Sugar on Me
1988	Taylor Dayne	I'll Always Love You
1987	The Bangles	Walk Like an Egyptian
1987	Heart	Alone
1987	Billy Idol	Mony Mony
1987	Billy Vera and the Beaters	At This Moment
1986	Dionne and Friends	That's What Friends Are For
1986	Lionel Richie	Say You, Say Me
1986	Prince and The Revolution	Kiss
1986	Steve Winwood	Higher Love
1985	Wham!	Careless Whisper
1985	Madonna	Like a Virgin
1985	Glenn Frey	The Heat Is On
1985	USA for Africa	We Are the World
1984	Prince	When Doves Cry
1984	Tina Turner	What's Love Got to Do with It
1984	The Romantics	Talking in Your Sleep
1984	Laura Branigan	Self Control
1983	The Police	Every Breath You Take
1983	Michael Jackson	Billie Jean
1983	Golden Earring	Twilight Zone
1983	Frida	I Know There's Something Going On
1982	Olivia Newton-John	Physical
1982	Survivor	Eye of the Tiger
1982	Foreigner	Waiting for a Girl Like You
1982	Rick Springfield	Don't Talk to Strangers
1981	Kim Carnes	Bette Davis Eyes
1981	Diana Ross and Lionel Richie	Endless Love
1981	The Pointer Sisters	Slow Hand
1981	Climax Blues Band	I Love You
1980	Blondie	Call Me
1980	Pink Floyd	Another Brick in the Wall, Part II
1980	KC and the Sunshine Band	Please Don't Go
1980	Styx	Babe
1979	The Knack	My Sharona
1979	Donna Summer	Bad Girls
1979	The Doobie Brothers	What a Fool Believes
1979	Chic	Good Times
1978	Andy Gibb	Shadow Dancing
1978	Bee Gees	Night Fever
1978	Yvonne Elliman	If I Can't Have You
1978	Chic	Dance, Dance, Dance (Yowsah, Yowsah, Yowsah)
1977	Rod Stewart	Tonight's the Night (Gonna Be Alright)
1977	Andy Gibb	I Just Want to Be Your Everything
1977	Eagles	Hotel California
1977	Marvin Gaye	Got to Give It Up
1976	Wings	Silly Love Songs
1976	Elton John and Kiki Dee	Don't Go Breaking My Heart
1976	Dorothy Moore	Misty Blue
1976	The Sylvers	Boogie Fever
1975	Captain and Tennille	Love Will Keep Us Together
1975	Glen Campbell	Rhinestone Cowboy
1975	Tony Orlando and Dawn	He Don't Love You (Like I Love You)
1975	Janis Ian	At Seventeen
1974	Barbra Streisand	The Way We Were
1974	Terry Jacks	Seasons in the Sun
1974	Blue Magic	Sideshow
1974	Blue Swede	Hooked on a Feeling
1973	Tony Orlando and Dawn	Tie a Yellow Ribbon Round the Ole Oak Tree

(continued)

Year	Artist	Song Name
1973	Jim Croce	Bad, Bad Leroy Brown
1973	Stevie Wonder	You Are the Sunshine of My Life
1973	Cher	Half-Breed
1972	Roberta Flack	The First Time Ever I Saw Your Face
1972	Gilbert O'Sullivan	Alone Again (Naturally)
1972	The Staple Singers	I'll Take You There
1972	Michael Jackson	Ben
1971	Three Dog Night	Joy to the World
1971	Rod Stewart	Maggie May/Reason to Believe
1971	Lee Michaels	Do You Know What I Mean
1971	Joan Baez	The Night They Drove Old Dixie Down
1970	Simon and Garfunkel	Bridge Over Troubled Water
1970	The Carpenters	(They Long to Be) Close to You
1970	Sly and the Family Stone	Thank You (Falettinme Be Mice Elf Agin)
1970	Eric Burdon and War	Spill the Wine
1969	The Archies	Sugar, Sugar
1969	The 5th Dimension	Aquarius/Let the Sunshine In
1969	Creedence Clearwater Revival	Proud Mary
1969	Jr. Walker and The All Stars	What Does It Take (To Win Your Love)
1968	The Beatles	Hey Jude
1968	Paul Mauriat	Love is Blue
1968	The Grass Roots	Midnight Confessions
1968	Sly and the Family Stone	Dance to the Music
1967	Lulu	To Sir With Love
1967	The Box Tops	The Letter
1967	Sam and Dave	Soul Man
1967	The Association	Never My Love
1966	The Mamas and the Papas	California Dreamin'
1966	? and the Mysterians	96 Tears
1966	Lee Dorsey	Working In The Coal Mine
1966	The Mindbenders	A Groovy Kind of Love
1965	Sam the Sham and the Pharaohs	Wooly Bully
1965	Four Tops	I Can't Help Myself (Sugar Pie Honey Bunch)
1965	Herman's Hermits	Mrs. Brown, You've Got a Lovely Daughter
1965	The Supremes	Stop! In the Name of Love
1964	The Beatles	I Want to Hold Your Hand
1964	The Beatles	She Loves You
1964	The Ray Charles Singers	Love Me with All Your Heart (Cuando Calienta El Sol)
1964	The Drifters	Under the Boardwalk
1963	The Beach Boys	Surfin' U.S.A.
1963	Skeeter Davis	The End of the World
1963	Inez and Charlie Foxx	Mockingbird
1963	Little Peggy March	I Will Follow Him
1962	Acker Bilk	Stranger on the Shore
1962	Ray Charles	I Can't Stop Loving You
1962	Mary Wells	The One Who Really Loves You
1962	Elvis Presley	Good Luck Charm
1961	Bobby Lewis	Tossin' and Turnin'
1961	Patsy Cline	I Fall to Pieces
1961	Ray Charles	Hit the Road Jack
1961	Sue Thompson	Sad Movies (Make Me Cry)
1960	Percy Faith	Theme from A Summer Place
1960	Jim Reeves	He'll Have to Go
1960	Brian Hyland	Itsy Bitsy Teenie Weenie Yellow Polka Dot Bikini
1960	Roy Orbison	Only the Lonely
1959	Johnny Horton	The Battle of New Orleans
1959	Bobby Darin	Mack the Knife
1959	Carl Dobkins Jr.	My Heart Is an Open Book
1959	The Everly Brothers	(Till) I Kissed You
1958	Domenico Modugno	Volare (Nel blu dipinto di blu)

(continued)

Year	Artist	Song Name
1958	The Everly Brothers	All I Have to Do Is Dream / Claudette
1958	Jimmie Rodgers	Secretly
1958	Danny and the Juniors	At the Hop
1957	Elvis Presley	All Shook Up
1957	Pat Boone	Love Letters in the Sand
1957	The Everly Brothers	Wake Up Little Susie
1957	Sam Cooke	You Send Me
1956	Elvis Presley	Heartbreak Hotel
1956	Elvis Presley	Don't Be Cruel
1956	Jim Lowe	Green Door
1956	The Four Lads	No, Not Much
1955	Perez Prado	Cherry Pink And Apple Blossom White
1955	Bill Haley and His Comets	Rock Around the Clock
1955	Joan Weber	Let Me Go, Lover!
1955	Nat King Cole	A Blossom Fell
1954	Kitty Kallen	Little Things Mean a Lot
1954	Perry Como	Wanted
1954	Tony Bennett	Stranger in Paradise
1954	Doris Day	If I Give My Heart to You
1953	Percy Faith	The Song from Moulin Rouge
1953	Les Paul and Mary Ford	Vaya con Dios
1953	Tony Bennett	Rags to Riches
1953	Silvana Mangano	Anna
1952	Leroy Anderson	Blue Tango
1952	Kay Starr	Wheel of Fortune
1952	Frankie Laine	High Noon
1952	Eddie Fisher	I'm Yours
1951	Nat King Cole	Too Young
1951	Tony Bennett	Because of You
1951	Les Paul and Mary Ford	The World Is Waiting for the Sunrise
1951	Guy Mitchell and Mitch Miller	My Truly, Truly Fair

Table S10.

The 276 song excerpts featured in the popular music corpus.

Factor	Group	Prop. of Variance
joyful activation	Listener	0.20
	Song	0.13
	Country	0.01
nostalgia	Listener	0.19
	Song	0.09
	Country	0.01
peacefulness	Listener	0.19
	Song	0.17
	Country	0.01
power	Listener	0.21
	Song	0.12
	Country	0.01
sadness	Listener	0.17
	Song	0.13
	Country	0.01
tenderness	Listener	0.20
	Song	0.14
	Country	0.01
tension	Listener	0.20
	Song	0.03
	Country	0.03
transcendence	Listener	0.20
	Song	0.02
	Country	0.02
wonder	Listener	0.24
	Song	0.04
	Country	0.02

Table S11.

Proportion of variance explained by the country the listener lives in, the song they heard, and the listener's own unique response pattern in response to popular music. These models use raw data aggregated to produce the nine factors reported in Zentner and colleagues'[5] work, which resulted in nine dimensions.

Factor	Group	Prop. of Variance
joyful activation	Listener	0.20
	Song	0.20
	Country	0.01
nostalgia	Listener	0.24
	Song	0.08
	Country	0.01
peacefulness	Listener	0.21
	Song	0.13
	Country	0.01
power	Listener	0.21
	Song	0.18
	Country	0.01
sadness	Listener	0.20
	Song	0.07
	Country	0.01
tenderness	Listener	0.21
	Song	0.11
	Country	0.01
tension	Listener	0.22
	Song	0.07
	Country	0.00
transcendence	Listener	0.20
	Song	0.02
	Country	0.00
wonder	Listener	0.27
	Song	0.06
	Country	0.00

Table S12.

Proportion of variance explained by the country the listener lives in, the song they heard, and the listener's own unique response pattern in response to unfamiliar music. These models use raw data aggregated to produce the nine factors reported in Zentner and colleagues'[5] work, which resulted in nine dimensions.

Factor	Group	Prop. of Variance
joyful activation	Listener	0.20
	Song	0.13
	Language	0.01
nostalgia	Listener	0.19
	Song	0.09
	Language	0.01
peacefulness	Listener	0.19
	Song	0.17
	Language	0.01
power	Listener	0.20
	Song	0.13
	Language	0.01
sadness	Listener	0.17
	Song	0.13
	Language	0.00
tenderness	Listener	0.20
	Song	0.14
	Language	0.01
tension	Listener	0.21
	Song	0.03
	Language	0.02
transcendence	Listener	0.20
	Song	0.02
	Language	0.02
wonder	Listener	0.24
	Song	0.04
	Language	0.02

Table S13.

Proportion of variance explained by the listener's native language, the song they heard, and the listener's own unique response pattern in response to popular music. These models use raw data aggregated to produce the nine factors reported in Zentner and colleagues' work[5], which resulted in nine dimensions.

Factor	Group	Prop. of Variance
joyful activation	Listener	0.20
	Song	0.13
	Language Family	0.01
nostalgia	Listener	0.19
	Song	0.09
	Language Family	0.01
peacefulness	Listener	0.20
	Song	0.17
	Language Family	0.01
power	Listener	0.20
	Song	0.13
	Language Family	0.01
sadness	Listener	0.17
	Song	0.13
	Language Family	0.01
tenderness	Listener	0.20
	Song	0.14
	Language Family	0.02
tension	Listener	0.22
	Song	0.03
	Language Family	0.03
transcendence	Listener	0.20
	Song	0.02
	Language Family	0.03
wonder	Listener	0.24
	Song	0.04
	Language Family	0.03

Table S14.

Proportion of variance explained by the language family that the listener's native language belonged to, the song they heard, and the listener's own unique response pattern in response to popular music. These models use raw data aggregated to produce the nine factors reported in Zentner and colleagues'[5] work, which resulted in nine dimensions.

Factor	Group	Prop. of Variance
joyful activation	Listener	0.20
	Song	0.20
	Language	0.01
nostalgia	Listener	0.24
	Song	0.08
	Language	0.00
peacefulness	Listener	0.21
	Song	0.13
	Language	0.00
power	Listener	0.21
	Song	0.18
	Language	0.01
sadness	Listener	0.20
	Song	0.07
	Language	0.01
tenderness	Listener	0.21
	Song	0.11
	Language	0.01
tension	Listener	0.22
	Song	0.07
	Language	0.00
transcendence	Listener	0.20
	Song	0.02
	Language	0.00
wonder	Listener	0.27
	Song	0.06
	Language	0.00

Table S15.

Proportion of variance explained by the listener's native language, the song they heard, and the listener's own unique response pattern in response to unfamiliar music. These models use raw data aggregated to produce the nine factors reported in Zentner and colleagues' work[5], which resulted in nine dimensions.

Factor	Group	Prop. of Variance
joyful activation	Listener	0.20
	Song	0.20
	Language Family	0.00
nostalgia	Listener	0.24
	Song	0.08
	Language Family	0.00
peacefulness	Listener	0.22
	Song	0.13
	Language Family	0.00
power	Listener	0.21
	Song	0.18
	Language Family	0.01
sadness	Listener	0.20
	Song	0.07
	Language Family	0.01
tenderness	Listener	0.21
	Song	0.11
	Language Family	0.01
tension	Listener	0.22
	Song	0.07
	Language Family	0.01
transcendence	Listener	0.20
	Song	0.02
	Language Family	0.00
wonder	Listener	0.27
	Song	0.06
	Language Family	0.00

Table S16.

Proportion of variance explained by the language family that the listener's native language belonged to, the song they heard, and the listener's own unique response pattern in response to unfamiliar music. These models use raw data aggregated to produce the nine factors reported in Zentner and colleagues'[5] work, which resulted in nine dimensions.

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